

Krish Tandon

PROJECT 0 · CS180/280A · FALL 2026

Becoming Friends with *Your Camera*

Three small experiments in the same idea: *where you stand* decides how a scene looks. Focal length only decides how much of it you keep.

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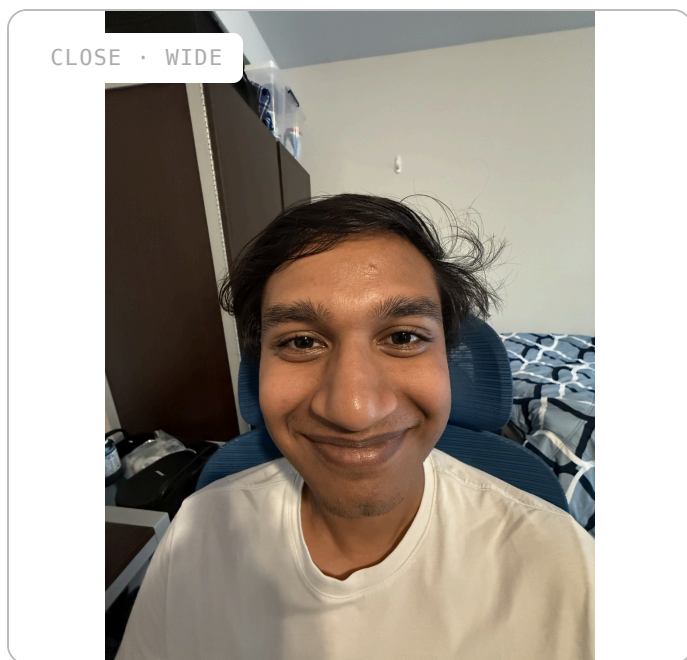
3 parts · 8 photos

iPhone 16 Pro

PART 1

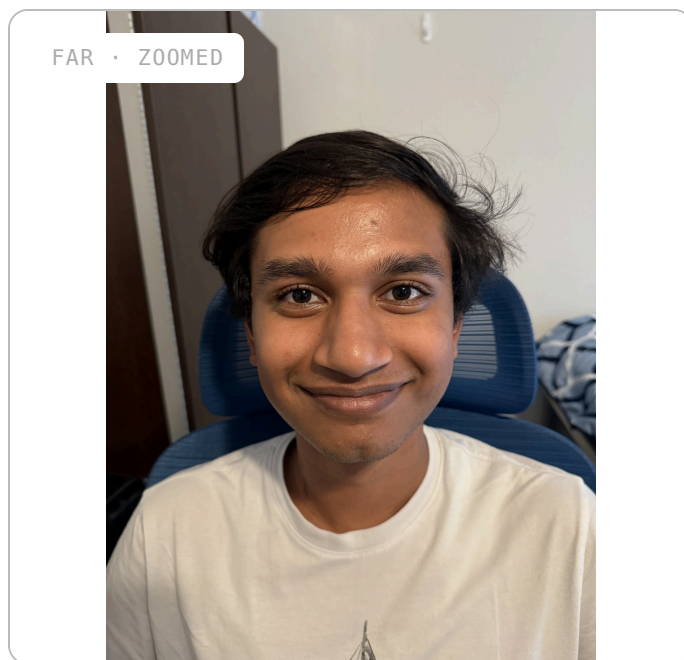
Selfie: the wrong way vs. the right way

Two portraits of the same face, framed to the same size. The first is a close selfie on the 0.5× ultra-wide. For the second I backed away and zoomed to 1× until the face filled the frame again. Nothing else changed — same subject, same light, same pose.



The wrong way. Shot close, on the 0.5× ultra-wide. The nose and forehead push forward, the jaw tapers away, and the ears wrap around out of sight.

14mm equiv. · f/2.2 · 1/60s · ISO 1000 · iPhone 16 Pro



The right way. Backed away and zoomed to 1× to rebuild the same framing. The nose settles back, the cheeks and jaw fill out, and the proportions read like the actual person.

24mm equiv. · f/1.8 · 1/60s · ISO 500 · iPhone 16 Pro



The same crop box from both frames, at the same scale — 0.5× on the left, 1× on the right. The head is the same height in each, so what you're seeing is purely the change in **proportions**: on the left the nose is larger, the jaw narrows, and the sides of the head fall away. On the right the face is fuller and flatter.

Why

In a perspective image, size falls off as **1/depth** — so what matters isn't how far away the camera is, it's the *ratio* of distances to the near and far parts of the face. A head is roughly 10 cm deep, so the nose sits at d while the ears sit at $d + \Delta$:

$$\text{nose} / \text{ears} = (d + \Delta) / d = 1 + \Delta/d$$

How far back did I actually go? I didn't measure it at the time — but the two photos do. Matching features across the pair, the face comes out the **same size in both** (0.98×) while the lens went from 14 mm to 24 mm equivalent — **1.71× longer**. Holding the subject's size while lengthening the lens by 1.71× is only possible if the camera moved **≈1.7× further away**.

Put numbers on it: from a typical arm's length of ~35 cm, that 10 cm of head depth renders the nose about **29%** bigger than the ears. At 1.7× that distance it's **16%**. The distortion roughly halves — which is exactly the difference you can see in the crop above.

So the zoom didn't fix the face — **moving back did**. Zooming only restored the framing. That's also why "85 mm is a flattering portrait lens" is a bit of a lie: the lens is flattering because it forces you to stand far enough away.

PART 2

Architectural perspective compression

The same experiment run backwards, on a building instead of a face. First I stood far away across an open space and zoomed all the way in. Then I walked toward the building and shot it wide, stopping when the facade filled roughly the same fraction of the frame.

FAR · ZOOMED IN



Flattened. Shot from far back at long focal length. The near and far ends of the facade are nearly the same size, the receding lines barely converge, and the building collapses toward a flat elevation drawing.

48mm equiv. · f/1.8 · 1/2273s · ISO 64 · iPhone 16 Pro

CLOSE · WIDE



Deep. Same building, walked right up to it, no zoom. The near corner looms, the far end races away, and the edges converge hard on the vanishing point.

24mm equiv. · f/1.8 · 1/2801s · ISO 80 · iPhone 16 Pro

Why

Same arithmetic, bigger scale. If the facade runs 20 m away from me in depth, then from 100 m back the far corner is only 17% smaller than the near one — the wall looks almost flat. Walk in to 15 m and the far corner drops to 43% of the near one, and the lines converge hard.

$$\text{far} / \text{near} = d / (d + L)$$

So the telephoto didn't compress anything — **distance did**. The long lens just let me stand that far back and still fill the frame.

PART 3

The dolly zoom

Parts 1 and 2 change the viewpoint and let the subject size change. The **dolly zoom** — Hitchcock's *Vertigo* shot — does the opposite: it changes the viewpoint and zooms at

the same time, *pinning the subject to a fixed size* so that the only thing left moving is the background.

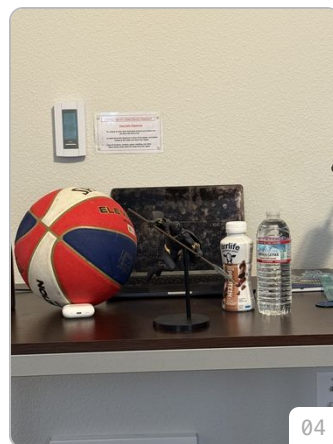
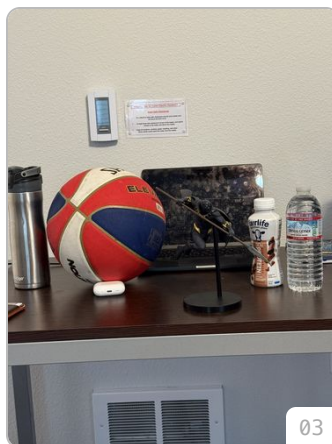
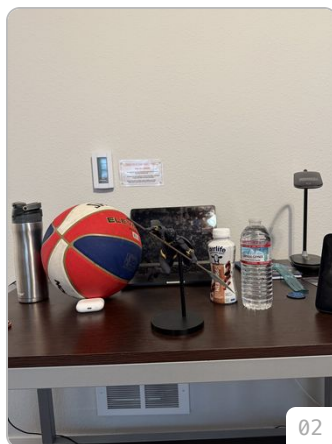
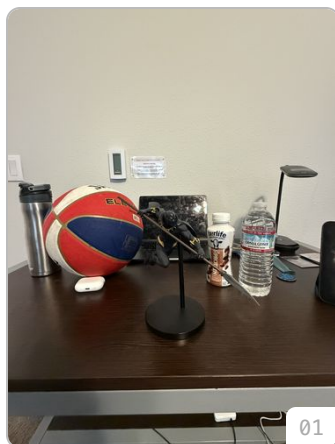
I shot four stills, stepping the camera back and zooming in between each one to hold the figure at the same size — 14 mm, 24 mm, 48 mm, then 72 mm equivalent — and assembled them into a GIF.



Subject locked. Background breathing. Tracking the figure and the safety sign separately across the four frames: with the figure held at a constant size, the wall behind it scales $1.00 \rightarrow 1.57 \rightarrow 2.12 \rightarrow 2.35\times$.

The frames

The four stills exactly as shot, in order — click any frame for the full-size image. Holding a subject steady by eye while walking backwards is harder than it sounds, so the figure drifts around a little here; the GIF above is these same four frames aligned on it. Aligning rescales each frame as a whole, so it removes my aim without touching the effect.



Why

Keeping the subject the same size means the focal length has to track the subject distance exactly ($f = k \cdot d_s$). The background sits a fixed distance D further back, so its magnification is:

$$m_{bg} \propto f / (d_s + D) = k / (1 + D/d_s)$$

As I dolly **back**, d_s grows, D/d_s shrinks, and the background **grows** — rushing in behind a subject that never changes size. Same fact as Parts 1 and 2, just made visible as motion: perspective is set by where the camera *is*, and the zoom is only there to hold the subject still so you can see it happen.

Takeaway

Zooming is just a crop — it scales everything by the same factor and can't change the relative size of two things at different depths. **Only moving the camera does that**, and that's all "perspective" really is.

A bulbous selfie nose, a flattened building, and the *Vertigo* shot are the same equation read three ways. The version I'll actually remember: **pick the distance first for the look you want, then pick the focal length to fix the framing.**